

1 Process Data Protocol

Overview

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The *Process data protocol* is used for the process analysis in the quality management and in the process data processing.

Purpose

The process data protocol is a summary of recorded values to get a quick overview during the process analysis.

The table view of the detail application offers an overview of existing entries. The system sorts the displayed information using table functions and complying with specified selection parameters. Each user can change the display of fields with the context menu.

Integration

The function displays recorded measured values of process characteristics in aggregated form based on selection criteria. These values must have been previously recorded in the system.

Selection parameters

The application provides the following selection criteria:

Type of evaluation

The evaluation type defines the relating object for the summary or if the user requires a calculation for single values for all process parameters. The following relating objects can be selected:

- Order
- OP (operation)
- Machine + shift date
- Machine + shift date + shift number

Machine

The system offers a detailed search for the machine using the pool application.

Evaluation period from - to

Specifies the time interval to be selected.

Order

The system offers a detailed search for the order using the pool application. This field is mandatory if the user selects the evaluation type "Order" or "Operation".

Operation

The system offers a detailed search for the operation using the pool application. This field is mandatory if the user selects the evaluation type "OP".

Include long-term data

Long-term data is included.

Please note that the system keeps all data used for these evaluations/reports in the server memory. Consequently, we do not recommend to select long periods of time and large data sets.

Technical background info:

Every time the system requests data in the client applications, it stores this data in the server memory and only then transfers the data to the client. If you require more memory space, you need to increase the server memory reserved for the Java application accordingly. Please refer to the MPDV support.

Detail application "Process data protocol"

The tabular report *process data protocol* displays the process data recorded and saved in the database including the following information: Please note that not all columns are filled with information depending on the selected evaluation type.

Category *Statistic parameter***Process parameter**

Technical name of the recorded process parameter

Mean value (AVG - order sequencing list)

Mean value of the recorded process parameter values. If the value exceeds the specification limit, the system highlights the field accordingly.

Maximum value

Maximum value of the recorded process parameter values. If the value exceeds the specification limit, the system highlights the field accordingly.

Minimum value

Minimum value of the recorded parameter values. If the value exceeds the specification limit, the system highlights the field accordingly.

Number of measured values

Number of recorded measured values meeting the selection criteria for the process parameter.

Category Order**Order**

Order number for which the process parameter was entered.

Article number of order

Stored article number of the order.

Article name of order

Stored article name of the order.

Category "Operation"**Operation**

MES order number for which the process parameter was recorded.

OP

OP number for which the process parameter was recorded.

OP name/designation

Stored OP designation of the operation

Article number of OP

Stored article name of the operation.

Article name of OP

Stored article name of the operation.

Category "Production parameter"**Machine**

Workplace/machine number indicating where the process parameter values were recorded.

Shift date

Shift date indicating when the process parameter values were recorded.

Shift number

Shift number of the recorded process parameter values.

Category "Primary quantities"**Yield (P)**

Recorded yield (primary) of the OP/machine

Scrap (P)

Recorded scrap quantity (primary) of the OP/machine

Quantity unit (P)

Quantity unit of the recorded quantity (primary)

Actual times category**Runtime**

Runtime of the OP/machine

Time of production

Production time of the OP/machine

Downtime

Downtimes of the OP/machine

Category "Specification"**Target value (TV)**

The target value for the collected process parameter.

Upper TL

The upper tolerance limit defined for this recorded process parameter.

UPAL (upper process action limit)

The upper process action limit defined for this recorded process parameter.

Lower TL

The lower tolerance limit defined for this recorded process parameter.

LPAL (lower process action limit)

The lower process action limit defined for this recorded process parameter.

Distribution

Graphical display of the distribution of measured values

Cp

The process capability index C_p is shown in the column C_p . The calculation of the process capability index C_p is based on the upper and lower specification limit and the corresponding standard deviation.

CpK

The process capability index C_{pK} is shown in the column C_{pK} . The calculation of the process capability index C_{pK} is based on the mean value, the corresponding standard deviation and the upper or lower specification limit. A high value shows that the production lies solidly within the specification limits.

Process location

Graphic display of information on mean value, minimum and maximum value and UTL and LTL.